

Scalability & Future Plan

Date	08 July 2026
Team ID	XXXXXX
Project Name	HDI Predictor using Machine Learning
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only limited HDI input parameters	Prediction accuracy depends on available features	High
2	Local deployment only	Cannot be accessed by users over the internet	Medium
3	No user authentication or data storage	User history and security features are unavailable	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small number of users	Deploy on cloud (Render/AWS/Azure) with multiple concurrent users
2	Data Storage	Uses static CSV dataset	Integrate MySQL or PostgreSQL database
3	Performance	Runs on a local Flask server	Optimize model and deploy using Gunicorn/Nginx
4	Security	Basic input validation	Add user authentication, HTTPS and secure API access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud Deployment (Render/AWS)	1 Month	Public access to the application
Phase 2	Database Integration	2 Months	Store user inputs and prediction history
Phase 3	Advanced Machine Learning Model	3 Months	Improve prediction accuracy
Phase 4	Mobile Application & Dashboard	4-6 Months	Better user experience and accessibility